

THE PROJECT

The "All Things" DevOps Chronicles

Or: How I Learned to Stop Worrying and Love the Pipeline

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Project: Todo API (The Ferrari Edition)

The Executive Summary

Not Just "Hello World"

Anyone can write a Todo app. This project is different. It is a **battle-tested, security-hardened, observable, containerized** microservice.

I didn't just "write code." I built an ecosystem.



DevOps Excellence

From Python to Kubernetes, every layer is optimized. It's designed to be safer than a bank account.

The Architecture

The Big Brain Stuff

Before diving into the details, this is how the data flows.

- **Focused Flow:** Not a spaghetti mess.
- **Laser Beam:** Requests travel cleanly from the user to the database.
- **Components:** FastAPI, Docker, K8s, and Monitoring all working in harmony.



Tech Stack: FastAPI (Python)



Clean Code

Chosen for simplicity.

Wrapped in **FastAPI** for speed and relying on **Pydantic** for robust data validation.



Data Validation

If garbage data is sent, the framework rejects it immediately. No need for 50 if/else statements.

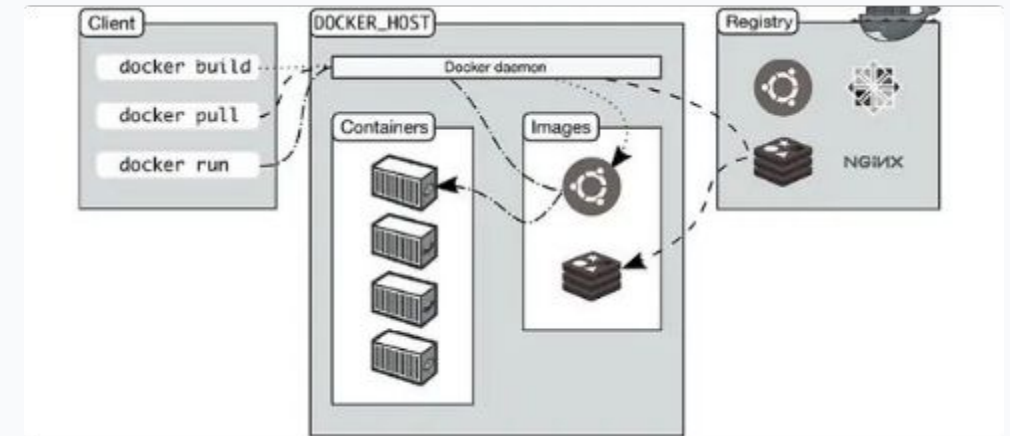


Middleware

Every request gets a X-Request-ID UUID. This allows for precise detective-style tracing if anything crashes.

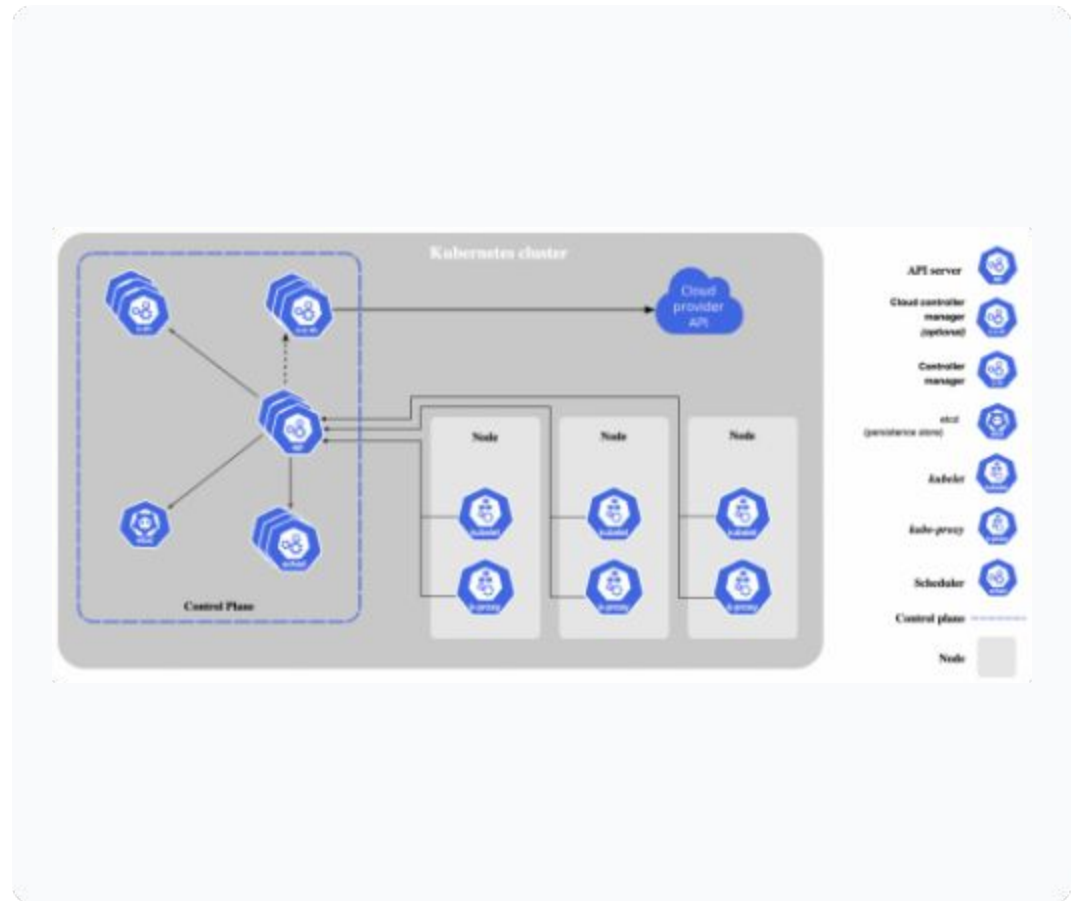
Docker: Not Just a VM

- **Multi-Stage Builds:** Uses the slim Python image to keep the footprint tiny.
- **Smart Caching:** requirements.txt is copied *before* the code. Docker skips the heavy pip install if dependencies haven't changed.
- **UnionFS:** Layers are stacked like a cake. Read-only bottom layers, thin Read-Write layer on top.
- **PID 1:** The app thinks it is the only process in the world.

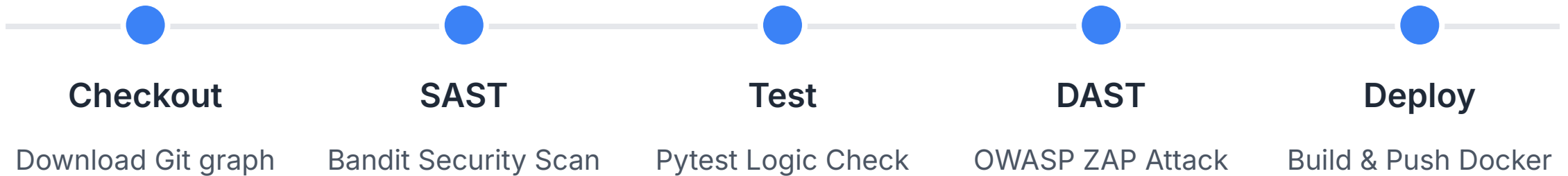


Kubernetes: The Conductor

- **ReplicaSet:** "I want 2 copies." K8s ensures 2 copies exist. If one dies, it respawns instantly.
- **The Service:** Pods change IPs constantly. The Service provides a stable "VIP" entrance for traffic.
- **Self-Healing:** It is impossible to kill this app. The system automatically repairs broken loops.



CI/CD: The Robot Butler



"I hate doing things manually, so I built a pipeline."

Security: Zero to Hero

The Scan Failed (Initially)

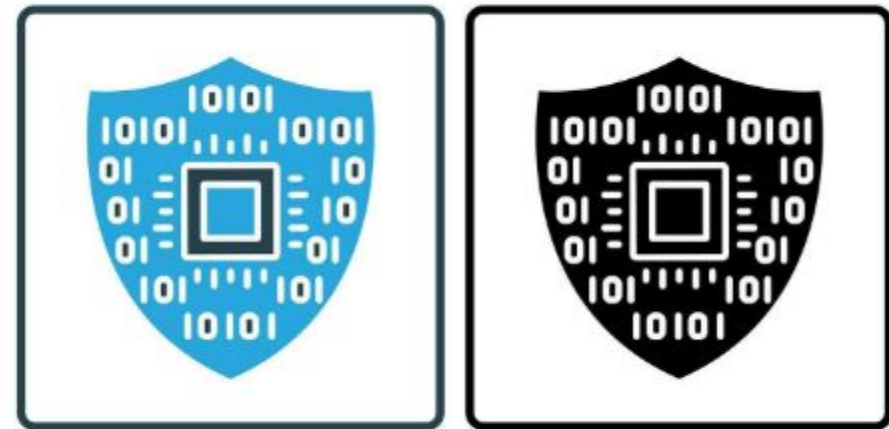
ZAP warned about missing isolation and Spectre vulnerabilities.

The Fix

I injected hardened headers in main.py:

- X-Content-Type-Options: nosniff
- Cross-Origin-Opener-Policy: same-origin
- Cache-Control: no-store

Result: A fully Green pipeline.



Observability: I See Everything



Metrics

Prometheus scrapes the /metrics endpoint. It returns raw data, not JSON:

```
# HELP http_requests_total
# TYPE counter
http_requests_total{status="2xx"} 4.0
http_requests_total{status="4xx"} 1.0
# HELP process_virtual_memory
process_virtual_memory_bytes 1.3e+08
```



Logs

Logs *are* structured JSON for machine indexing:

```
{
  "level": "INFO",
  "msg": "User login",
  "request_id": "abc-123"
}
```



Tracing

The X-Request-ID ties it all together. I can trace a specific ID through the logs and metrics to find issues instantly.

Why 20/20?

It Scales. It Defends. It Informs.

This project is the culmination of everything we learned, wrapped in a 150-line backend that hits every note.

Git commit. Push. Drop mic. 🎤